Ji 2020 cutaneous SCC — analysis summary

RUN 8 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Malignant cells from 8/10 samples (≥100 cells) form one continuous, sample-mixed manifold (iLISI≈4.0; single kNN component; 13 detached cells removed). Cell-cycle regression was required — it is the largest program and otherwise dominates the embedding.

Gate 2 — module geography → GO (branching)

The embryonic modules resolve into a branching wishbone: a Gast/pSom co-expression hub diverges into a Gast-leaning (Basal) arm and a pSom-leaning (Differentiated) arm. Crest is a terminal readout at the tips, uncoupled from Gast/pSom.

Gate 3 — hub cross-repression → GO (branching)

Within the 4,888-cell Mixed hub, per-gene Gast and pSom loadings mutually anti-correlate: r = −0.54 (Puram anchor ≈ −0.49) — competitive cross-repression, the bifurcation signature.

STEP FIGURES

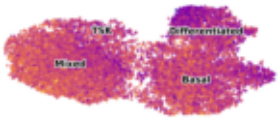
Manifold — samples



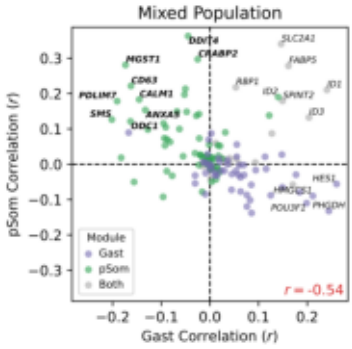
Trajectory / root



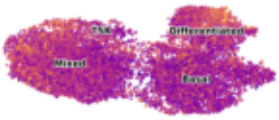
Gastrulation



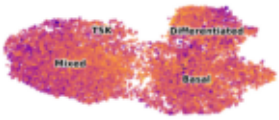
Gate 3: r = −0.54



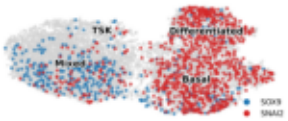
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10 patients, 48,164 cells; 10,598 malignant → 10,518 after QC. Subtypes Mixed / Basal / Differentiated / TSK.

Verdict.
Pattern reproduces — all three gates GO. Branching wishbone rooted at the Gast/pSom hub; competitive Gast↔pSom cross-repression (r=−0.54); Crest terminal, uncoupled. Gate 1 GO / Gate 2 GO / Gate 3 GO. EMT-TFs SOX9 (hub/basal), SNAI2 (Differentiated arm).

Parameter choices

Min cells/sample	≥100 (gap 40→268, 6.7×; keeps 8/10, drops P3, P5)
Island removal	UMAP radius-graph, largest component (−13 cells)
Embedding	UMAP min_dist = 0.4, spread = 1.0
Geometry / root	Branching (3-tip wishbone); root = Mixed hub (node 16, a tip)
EMT-TFs	SOX9 (hub/basal), SNAI2 (Differentiated arm) — Snail/Zeb/Twist ~absent
Cell-cycle regression	S + G2M scores regressed — KEY to a clean manifold
Integration	Harmony on sample, θ = 2, 20 it; n_pcs = 20
Trajectory	scFates PPT tree, Nodes = 60, λ = 2000, σ = 1.5
Hub / Gate-3 r	Mixed (4,888 cells); r = −0.54